1. $\dfrac{m_0c^2}{\sqrt{1-(\dfrac{v}{c})^2}} = km_0c^2$ $\qquad v = \dfrac{c}{k}\sqrt{k^2-1}$

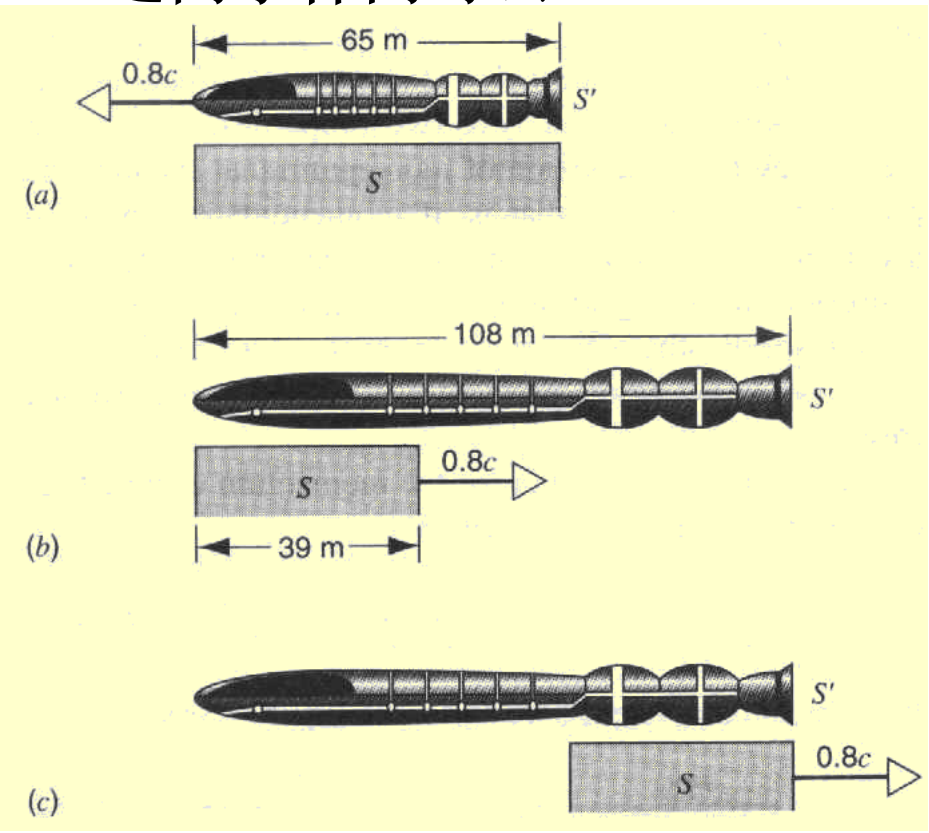
$$E_k = mc^2 - m_0c^2 = (k-1)m_0c^2$$

2. $$-\bar{F}\Delta l = (m_2c^2 - m_0c^2) - (m_1c^2 - m_0c^2) = m_2c^2 - m_1c^2$$

$$\bar{F} = 10^{-10}\,N$$

3. $$\Delta x = \frac{\Delta x' + u\Delta t'}{\sqrt{1 - \frac{u^2}{c^2}}} \qquad \Delta x' = 80m \qquad \Delta t' = \frac{80}{c} \qquad \Delta x = 240m$$

观察者S站在长度65m的站台上。火箭以0.8c平行站台运动，某一瞬间观察者S看到火箭的头和尾正好与站台两端重合。求：(a) S认为火箭通过站台上某一点所需时间；(b)火箭的固有长；(c)火箭上的观察者S′认为站台的长度是多少；(d)S′认为S通过火箭所需时间；(e)S认为火箭的两端同时与站台两端重合，S′认为这两事件同时么？



(a) $\Delta t_0 = L/0.80c = 0.27\mu s$

(固有时)

(b) $L_0 = \gamma L = 108\text{m}$

(固有长)

(c) $D = D_0/\gamma = 39\text{m}$

(d) $\Delta t' = 108/0.80c = 0.45\mu s$

(e) $\Delta t' = \gamma(\Delta t - V\Delta x/c^2)$

$= -\gamma V\Delta x/c^2 = 0.29\mu s$